The systems approach & control

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State the type of loop used in the flowchart.

- Fixed loop

No marks for: Open loop, closed loop, continuous loop

3 Quick check

- a) Open loop
- b) Process
- c) Temperature sensor

4 Key terms

- 1 – C
- 2 – A
- 3 – D
- 4 – E
- 5 – B

5 State whether the dryer uses open loop or closed loop control, and justify your answer.

- Open loop
- There is no feedback: it runs for a set time whether the hands are dry or not

6 Describe how the closed loop control system keeps the temperature at 22 °C.

- The thermistor (sensor) measures the actual temperature and feeds it back
- The controller/comparator compares the actual temperature with the set level of 22 °C
- If it is too low the heater is switched on; when 22 °C is reached it switches off

No marks for: "It uses feedback" with no sensor or comparison

7 Draw a sub-system diagram for the barrier, including feedback.

- Car sensor input to the controller
- Controller output to motor driver/motor
- Barrier as the output (rotary motion)
- Up and down micro switches feeding back to the controller with arrows in the right direction

No marks for: Leaving out the feedback switches